

PRAKHAR SHRIVASTAVA

Mechanical Engineering | CAD, CAE & Design Validation

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PROFILE

Final-year Mechanical Engineering student (graduating 2027, available immediately for full-time GET roles) with hands-on experience across CAD modelling, multibody dynamics simulation, DFMEA/PFMEA, and manufacturing/process documentation gained through two industry internships and three years of core-team project work. Comfortable moving across design, analysis, and validation functions; specialised strength in vehicle dynamics and mechanism design, built through MSC Adams-based suspension, steering, and mechanism modelling.

TECHNICAL SKILLS

- **CAD & Design:** Creo Parametric, SolidWorks, Siemens NX, Solid Edge — part & assembly modelling, GD&T, mass/inertia extraction, BOM management
- **Simulation & Analysis:** MSC Adams (multibody dynamics), ANSYS (modal, random vibration, thermal, co-simulation), IPG CarMaker
- **Quality & Process:** DFMEA/PFMEA, MOST analysis, Lean Six Sigma exposure, drop-test and hardware correlation, BOM verification/consolidation
- **Domain Application:** Suspension/steering kinematics, mechanism design, gearbox/drivetrain analysis, jig & fixture design
- **Scripting & Documentation:** Python, C, MS Office; technical drawings and engineering documentation

WORK EXPERIENCE

R&D Intern — Avinya Miltech Pvt. Ltd., Indore

Jun 2026 – Jul 2026

- Built a kinematic and dynamic MBD model of a CMP (Chemical Mechanical Polishing) machine in MSC Adams, incorporating contact, friction, RPM, and trajectory optimisation.
- Redesigned a 4-bar lens-polishing mechanism to eliminate quick-return motion using constraint-based mechanism analysis.
- Ran modal, random vibration, and thermal analyses on an optomechanical assembly in ANSYS; produced GD&T drawings/BOM in Solid Edge and selected bearings to improve service life.
- Derived conic disc profiles from depth-of-cut data and built a Python/HTML tool to automate trajectory prediction and re-flattening (Preston's equation, Material Removal Rate).

NPD Intern — Gabriel India Ltd., Dewas

May 2025 – Jul 2025

- Studied twin-tube, mono-tube, and strut damper architectures/mounting systems for OEM applications; verified and consolidated BOMs across 1,000+ damper assemblies.
- Conducted DFMEA/PFMEA and applied Lean Six Sigma and MOST analysis to streamline assembly-line workflows.

Multibody Dynamics Head; Vehicle Dynamics Member — Team GS Racers, SGSITS Indore

2023 – Present

- Designed front double-wishbone and rear H-arm suspension in MSC Adams with full hardpoint definition and constraint equations; ran MBD simulations for camber/toe gain, roll centre, Ackermann %, and steering effort.
- Performed four-post rig, sinusoidal ride, and drop-test analyses to correlate simulation to hardware and extract ride frequencies and upright load cases; designed manufacturing jigs/fixtures.
- Led dynamic MBD analysis of the gearbox for torque transfer, backlash, and shaft alignment; validated CV joint safety via plunge/articulation angle analysis.
- As Multibody Dynamics Head, led the MBD division across suspension, steering, and powertrain analysis, supervising a cross-functional team and training juniors on Creo, SolidWorks, and MSC Adams.

TECHNICAL PROJECTS

Reaction Wheel Attitude Control System for Nanosatellites —

Jan – Mar 2026

- Modelled a reaction wheel assembly in Creo with mass balancing; simulated spin-up/attitude response in MSC Adams and built a 3-axis Euler control architecture, validated in ANSYS.

Rocker-Bogie Suspension for a Planetary Rover —

Jan – Feb 2026

- Modelled a rocker-bogie rover in SolidWorks with joints/constraints/contact in MSC Adams; ran rough-terrain ride analysis to evaluate articulation, stability, and load distribution.

ACHIEVEMENTS

- 1st Rank, Validation Event, E-BAJA 2026; 7th overall rank, M-BAJA 2024 (also competed in E-BAJA '24/'25).
- AIR 91488, JEE Mains 2023; School topper, Classes IX–XII.

EDUCATION

Degree	Institute	Year	CGPA
B.Tech, Mechanical Engineering	SGSITS, Indore	2027 (Exp.)	Pending
Class XII (CBSE)	Maharishi Vidya Mandir, Jabalpur	2022	90.0%
Class X (CBSE)	Maharishi Vidya Mandir, Jabalpur	2020	89.4%

